

Lecture 2 — Solving a Linear Program

Course notes: the graphical method, the corner method, and all the possible cases

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What you will learn. After this lecture you can solve any linear program with two variables, by drawing (graphical method) or by computing the corners (corner method), and you can recognise every case that can happen: one best plan, many best plans, no best plan, no feasible plan, and three special situations.

How to read these notes. Methods and key facts are in boxes. Every computation is shown step by step. The answers to "Check yourself" are at the end.

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1 A plan is a point of the plane

We use the drone workshop of Lecture 1:

$$\begin{array}{lll} \max & z = 40x_1 + 60x_2 & \text{(profit, thousand DA per week)} \\ \text{s.t.} & x_1 + 2x_2 \le 40 & \text{(assembly hours)} \\ & 3x_1 + 2x_2 \le 60 & \text{(testing hours)} \\ & x_2 \le 18 & \text{(cameras)} \\ & x_1, x_2 \ge 0 & \end{array}$$

With two variables, a plan (x_1, x_2) is a *point* of the plane. Each constraint keeps the points that are on one side of a line, and the *feasible set* is the region where all the constraints are satisfied. We can draw it.

2 Three tools

2.1 Tool 1: draw a line

The constraint $3x_1 + 2x_2 \le 60$ is limited by the line $3x_1 + 2x_2 = 60$. To draw a line, find two of its points. The easiest points are on the axes:

- put $x_1 = 0$: then $2x_2 = 60$, so $x_2 = 30$; this gives the point $(0, 30)$;
- put $x_2 = 0$: then $3x_1 = 60$, so $x_1 = 20$; this gives the point $(20, 0)$.

Join the two points with a ruler. Some lines are special:

- $x_2 = 18$ is a *horizontal* line (all points with second coordinate 18);
- $x_1 = 5$ is a *vertical* line (all points with first coordinate 5);
- $x_1 - x_2 = 0$ passes through the origin: both axis points are $(0, 0)$, so take another point, for example $(1, 1)$.

2.2 Tool 2: choose the correct side

The arrow of coefficients. For the constraint $3x_1 + 2x_2 \le 60$, take its coefficients 3 and 2 and draw the arrow $a = (3, 2)$ from a point of the line (Figure 1).

- The arrow is *perpendicular* to the line.
- In the direction of the arrow, the expression $3x_1 + 2x_2$ *grows*: at $(0,0)$ it is 0, at $(3,2)$ it is $9+4=13$.
- So a constraint "$\leq$" keeps the side *opposite* to the arrow, and a constraint "$\geq$" keeps the side of the arrow.

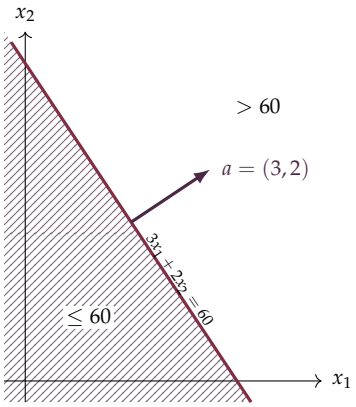


Figure 1: The arrow of coefficients $a = (3,2)$ is perpendicular to the line and points to where $3x_1 + 2x_2$ grows. The constraint $3x_1 + 2x_2 \leq 60$ keeps the hatched side.

The test point. To check (or instead of the arrow), take a *test point* that is not on the line, and check the constraint at this point:

- if the constraint is *true* at the test point, keep the side of the line that contains the test point;
- if it is *false*, keep the other side.

The origin $(0,0)$ is usually the easiest test point.

- $3x_1 + 2x_2 \leq 60$ at $(0,0)$: $0 \leq 60$ is true. Keep the side of the origin.
- $x_1 + x_2 \geq 6$ at $(0,0)$: $0 \geq 6$ is false. Keep the other side, away from the origin.
- $x_2 \leq 2x_1$: the origin is on the line, so it cannot be a test point. Use $(1,0)$: $0 \leq 2$ is true; keep the side of $(1,0)$.

2.3 Tool 3: compute a corner

Method 2.1 — Compute a corner

A corner is a point where two lines meet. Write the two equations and solve the system (by elimination or by substitution). Always check the result in both equations.

By elimination. Solve $x_1 + 2x_2 = 40$ (1) and $3x_1 + 2x_2 = 60$ (2). Both contain $2x_2$, so subtract: (2) $-$ (1) gives $2x_1 = 20$, so $x_1 = 10$. Then (1) gives $10 + 2x_2 = 40$, so $x_2 = 15$. Check in (2): $30 + 30 = 60$. The corner is $(10, 15)$.

By substitution. Solve $2x_1 + x_2 = 8$ and $x_1 + 2x_2 = 7$. From the first, $x_2 = 8 - 2x_1$. Put it in the second: $x_1 + 2(8 - 2x_1) = 7$, so $x_1 + 16 - 4x_1 = 7$, $-3x_1 = -9$, $x_1 = 3$, and $x_2 = 8 - 6 = 2$. Check: $6 + 2 = 8$ and $3 + 4 = 7$. The corner is $(3, 2)$.

Check yourself

Q1. Draw the line $x_1 + 3x_2 = 12$: give its two points on the axes. Which side does $x_1 + 3x_2 \leq 12$ keep?
Q2. Compute the point where the lines $2x_1 + x_2 = 10$ and $x_1 - x_2 = 2$ meet.

3 The graphical method on the drone workshop

Step 1: the feasible set. Draw the three lines: assembly through $(40, 0)$ and $(0, 20)$, testing through $(20, 0)$ and $(0, 30)$, cameras $x_2 = 18$ (horizontal). The origin satisfies the three constraints, so we keep the side of the origin each time. With $x_1, x_2 \geq 0$ the feasible set is the polygon O A B C D below, with

$$\mathrm{O} = (0,0), \quad \mathrm{A} = (20,0), \quad \mathrm{B} = (10,15), \quad \mathrm{C} = (4,18), \quad \mathrm{D} = (0,18).$$

A is on the testing line and the axis $x_2 = 0$; B is on the assembly and testing lines (computed above); C is on the assembly and camera lines ($x_2 = 18$, $x_1 + 36 = 40$, $x_1 = 4$); D is on the camera line and the axis $x_1 = 0$.

Figure 2 shows how the feasible set is built, one constraint at a time: each constraint keeps a half-plane, and the feasible set is the part common to all of them.

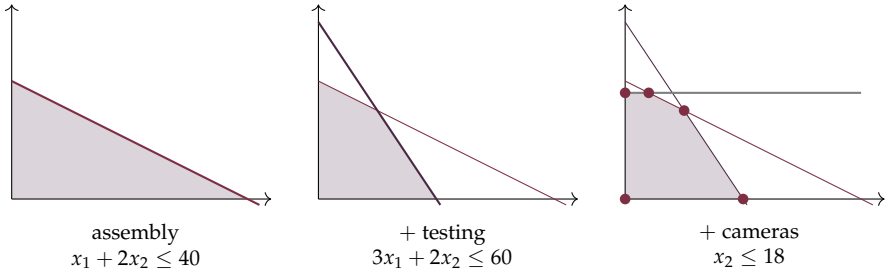


Figure 2: Building the feasible set of the drone workshop, one constraint at a time (with $x_1, x_2 \geq 0$).

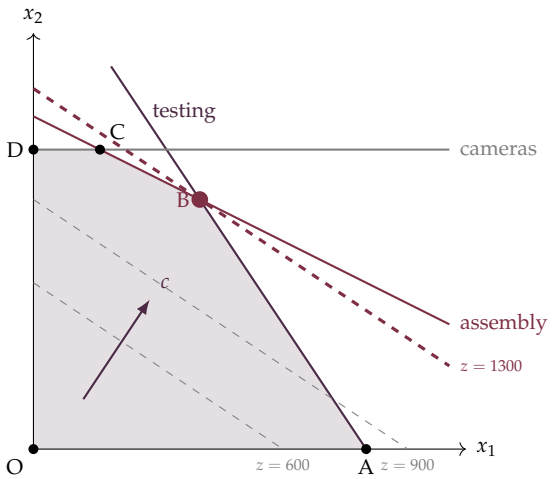


Figure 3: The drone workshop: feasible set O A B C D, lines of equal profit, and the optimal corner B.

Step 2: lines of equal profit. All the plans with the same profit k are on the line $40x_1 + 60x_2 = k$. For $k = 600$ the line passes through $(15, 0)$ and $(0, 10)$; for $k = 900$ through $(22.5, 0)$ and $(0, 15)$.

These lines are parallel (only k changes). When k grows, the line moves up and to the right, in the direction of the arrow $c = (40, 60)$ made of the coefficients of z.

Step 3: push the line. Slide the line parallel to itself in the direction of c, as far as possible, while it still touches the feasible set. The last point touched is the corner B $= (10, 15)$. Its profit is $40 \times 10 + 60 \times 15 = 1300$ thousand DA. A line with a larger profit does not touch the feasible set, so no feasible plan gives more than 1300 thousand DA.

Answer. The optimal plan is 10 drones A and 15 drones B per week; the optimal value is 1300 thousand DA. These are whole numbers: as explained in Lecture 1 (Section 2.7), this is luck, and here the best plan of the LP relaxation is also the best plan of the real problem.

Reading the solution.

Constraint	Used at B	Available	Unused (slack)	
Assembly	$10 + 30 = 40$	40	0	binding
Testing	$30 + 30 = 60$	60	0	binding
Cameras	15	18	3	not binding

A constraint is *binding* (or *active*) when it is satisfied with equality at the optimal plan: its line passes through the optimal point. Binding constraints are the bottlenecks: to earn more, the workshop needs more assembly or testing hours; more cameras would not help, since 3 cameras are already unused.

4 Why the best plan is at a corner

Key fact 2.1

If a linear program has an optimal solution, then at least one corner of the feasible set is an optimal solution.

4.1 The feasible set has no holes and no dents

If two plans P and Q are feasible, then every plan on the segment between them is also feasible. Indeed, take one constraint $a_1x_1 + a_2x_2 \leq b$. When we move from P to Q along the segment, the value of $a_1x_1 + a_2x_2$ changes regularly from its value at P to its value at Q (see the computation on an edge below). Both values are at most b, so every value in between is also at most b. A set with this property is called *convex* (Figure 4). The feasible set of a linear program is always convex.

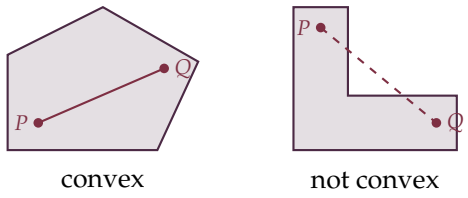


Figure 4: Left: a convex set, the segment between two of its points stays inside. Right: a set that is not convex. The feasible set of a linear program always looks like the left picture.

4.2 Sliding to a corner

Here is the idea of Key fact 2.1 on the drone workshop (Figure 5).

1. Start from any feasible plan, for example $(6, 6)$, with $z = 600$.
2. Move in the direction of the arrow $c = (40, 60)$, where the profit grows. With the direction $(2, 3)$ (the same direction as c), the points are $(6 + 2t,\ 6 + 3t)$. The first constraint to stop us is testing: $3(6 + 2t) + 2(6 + 3t) = 30 + 12t \leq 60$ gives $t \leq 2.5$ (assembly would allow $t \leq 2.75$ and cameras $t \leq 4$). We stop at $(11, 13.5)$ on the border, with $z = 440 + 810 = 1250$.
3. Now we are on an edge (the testing line). Along this edge, in one of the two directions the profit does not decrease; here the direction $(-2, 3)$ keeps $3x_1 + 2x_2 = 60$ and changes z by $40 \times (-2) + 60 \times 3 = 100 > 0$ per step. We move in this direction until the corner B $= (10, 15)$, with $z = 1300$.

So from any feasible plan we can reach a corner which is at least as good. This is why the best value is always reached at a corner.

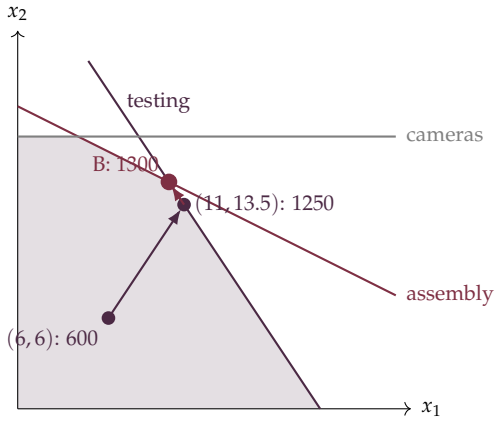


Figure 5: Sliding to a corner: from $(6, 6)$ in the direction of c to the border, then along the border to the corner B.

4.3 The algebra on an edge

Why does it always work? We explain it on the drone workshop, but the argument is general.

1. *On a segment, the value changes regularly.* Take the edge from C $= (4, 18)$ to B $= (10, 15)$. To go from C to B, we add 6 to x_1 and -3 to x_2. So the points of the edge are $(4 + 6t,\ 18 - 3t)$ with t from 0 (point C) to 1 (point B). Their profit is

$$40(4 + 6t) + 60(18 - 3t) = 160 + 240t + 1080 - 180t = 1240 + 60t.$$

   This is a straight-line function of t (like $at + b$). When t goes from 0 to 1, it goes regularly from 1240 to 1300. So on the edge, the best value is at an end: here at B. The same computation works on any segment: the value of z is always of the form $z(\text{first end}) + t \times (\text{difference})$.
2. *Every point of the polygon is on a segment between two points of the border* (draw a horizontal line through it). So its value is at most the value of one of these border points.
3. *Every point of the border is on an edge between two corners.* So its value is at most the value of one of these corners.

Together: no point of the polygon has a better value than the best corner.

Method 2.2 — The corner method

1. Draw the feasible set and compute all its corners (Method 2.1).
2. Compute z at each corner.
3. Choose the best corner.

Warning: if the feasible set is unbounded (it goes to infinity), first check that an optimal solution exists (see Cases 3 and 4).

For the drone workshop:

Corner	(x_1, x_2)	$z = 40x_1 + 60x_2$
O	$(0, 0)$	0
A	$(20, 0)$	800
B	$(10, 15)$	**1300**
C	$(4, 18)$	1240
D	$(0, 18)$	1080

The best corner is B, with 1300 thousand DA, as with the graphical method.

5 How can we be sure that a plan is optimal?

A picture is convincing, but here is a way to be completely sure, with simple algebra. *If we multiply constraints by positive numbers and add them, we get a new inequality that every feasible plan satisfies* (an inequality keeps its direction when we multiply it by a positive number, and we can add inequalities in the same direction).

Try 1. Multiply the assembly constraint by 40: $40x_1 + 80x_2 \leq 1600$. Since $x_2 \geq 0$, we have $60x_2 \leq 80x_2$, so for every feasible plan

$$40x_1 + 60x_2 \leq 40x_1 + 80x_2 \leq 1600.$$

No plan gives more than 1600 thousand DA. This is true, but not enough: no plan reaches 1600.

Try 2. Take 25 times the assembly constraint plus 5 times the testing constraint:

$$25(x_1 + 2x_2) + 5(3x_1 + 2x_2) \leq 25 \times 40 + 5 \times 60, \qquad \text{that is} \qquad 40x_1 + 60x_2 \leq 1300.$$

No plan gives more than 1300 thousand DA, and the plan B $= (10, 15)$ gives exactly 1300 thousand DA. So B is optimal: this is a complete proof.

How to find the numbers 25 and 5? Call them y_1 (for assembly) and y_2 (for testing), and ask the left side $y_1(x_1 + 2x_2) + y_2(3x_1 + 2x_2)$ to be exactly the profit $40x_1 + 60x_2$. Comparing the coefficients of x_1 and of x_2:

$$y_1 + 3y_2 = 40, \qquad 2y_1 + 2y_2 = 60.$$

The second gives $y_1 = 30 - y_2$; then $30 - y_2 + 3y_2 = 40$, so $y_2 = 5$ and $y_1 = 25$. Both are positive, as needed. We used the two constraints that are binding at B. (In Chapter 3 we will see that $y_1 = 25$ is the value of one more assembly hour, and $y_2 = 5$ the value of one more testing hour.)

6 The graphical method in general, and a minimisation

Method 2.3 — The graphical method

1. Draw each constraint line and keep the correct side (test point).
2. Shade the feasible set. If it is empty, the problem is *infeasible*: stop.
3. Draw one line of equal value $z = k$ and the arrow c of the coefficients of z.
4. Push the line in the direction of c for a max, or in the *opposite* direction for a min, as far as possible while it still touches the feasible set.
5. If it can be pushed forever, the problem is *unbounded*: stop.
6. Otherwise read the last corner (or edge) touched, compute it exactly with Method 2.1, and give z^*.

Example: spare-part boxes. A maintenance team buys boxes of spare parts for its robots. A box A contains 2 motors and 1 sensor and costs 30 thousand DA; a box B contains 1 motor and 2 sensors and costs 40 thousand DA. The team needs at least 8 motors and 7 sensors. With x_1, x_2 the numbers of boxes A and B:

$$\min\ z = 30x_1 + 40x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \geq 8, \quad x_1 + 2x_2 \geq 7, \quad x_1, x_2 \geq 0.$$

- *Lines.* $2x_1 + x_2 = 8$ through $(4,0)$ and $(0,8)$; $x_1 + 2x_2 = 7$ through $(7,0)$ and $(0,3.5)$.
- *Sides.* At $(0,0)$: $0 \geq 8$ is false and $0 \geq 7$ is false, so for both constraints we keep the side away from the origin. The feasible set goes to infinity (up and to the right).
- *Corners.* $(0,8)$; $(7,0)$; and the intersection of the two lines, $(3,2)$ (computed in Section 2.3).
- *Push.* For a minimisation, push the line $30x_1 + 40x_2 = k$ towards the origin. The last point touched is $(3,2)$.
- *Check with the corners.* $(0,8)$: 320; $(3,2)$: $90 + 80 = 170$; $(7,0)$: 210.

Answer. Buy 3 boxes A and 2 boxes B, for 170 thousand DA. Both constraints are binding: exactly 8 motors and 7 sensors.

Check yourself

Q5. Check that $\frac{20}{3} \times (2x_1 + x_2 \geq 8) + \frac{50}{3} \times (x_1 + 2x_2 \geq 7)$ gives $30x_1 + 40x_2 \geq 170$ (combining constraints, as in Section 5). What does it prove?

7 All the possible cases

Case	What we see	Answer
1. Unique optimum	the line touches one corner last	one optimal plan
2. Many optima	the line touches a whole edge last	infinitely many optimal plans
3. Unbounded set, optimum exists	the set goes to infinity, but the line stops	an optimal plan exists
4. Unbounded problem	the line can be pushed forever	no optimal plan
5. Infeasible problem	the feasible set is empty	no feasible plan

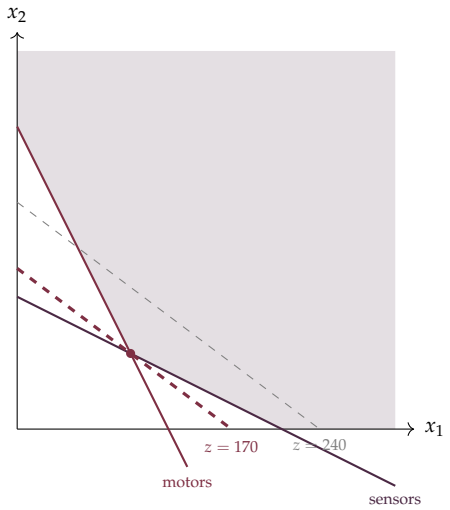


Figure 6: Spare-part boxes: the feasible set goes to infinity, and the cheapest plan is the corner $(3, 2)$.

7.1 Case 1: a unique optimal solution

The line of equal value touches only one corner last. Example: the drone workshop, optimum B $= (10, 15)$.

7.2 Case 2: infinitely many optimal solutions

Suppose the profit of a drone A is 30 thousand DA: $z = 30x_1 + 60x_2$. Since $30x_1 + 60x_2 = 30(x_1 + 2x_2)$, the lines of equal value are parallel to the assembly line $x_1 + 2x_2 = 40$. When we push them, the last contact is the whole edge from C to B. With the corner method: B gives $300 + 900 = 1200$ and C gives $120 + 1080 = 1200$: the same best value. All the points of the edge are optimal:

$$(4 + 6t,\ 18 - 3t), \quad 0 \le t \le 1, \qquad z = 1200.$$

For example $t = \frac{1}{2}$ gives $(7, 16.5)$ with $z = 210 + 990 = 1200$.

How to recognise it: two neighbouring corners give the same best value, or the objective is parallel to a binding constraint. *How to write the answer:* give the two corners and the formula of all the points between them.

7.3 Cases 3 and 4: an unbounded feasible set

Take the feasible set $x_1 - x_2 \le 1$, $x_1, x_2 \ge 0$. It goes to infinity.

Case 3: min $x_1 + x_2$**.** For every feasible plan, $x_1 \ge 0$ and $x_2 \ge 0$, so $x_1 + x_2 \ge 0$; and $(0, 0)$ is feasible with value 0. So $(0, 0)$ is optimal, $z^* = 0$. An unbounded feasible set *can* have an optimal solution.

Case 4: max $x_1 + x_2$**.** The line can be pushed forever. *Proof with a family of points:* for $t \ge 0$, the point $(1 + t,\ t)$ satisfies $(1 + t) - t = 1 \le 1$, $1 + t \ge 0$ and $t \ge 0$: it is feasible. Its value is $1 + 2t$, which becomes as large as we want when t grows. So no plan can be optimal: the problem is *unbounded*. In practice, this usually means that a constraint is missing in the model.

How to write the answer in Case 4: give a family of feasible points $x(t)$, $t \ge 0$, and show that $z(x(t)) \to +\infty$ (or $-\infty$ for a minimisation).

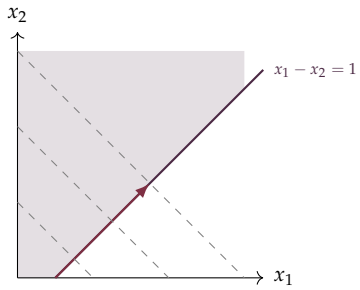


Figure 7: The feasible set $x_1 - x_2 \leq 1$, $x_1, x_2 \geq 0$ goes to infinity (Cases 3 and 4). The arrow shows the family of points $(1 + t, t)$.

7.4 Case 5: no feasible solution

max $x_1 + x_2$ s.t. $x_1 + x_2 \leq 1$, $x_1 + x_2 \geq 2$, $x \geq 0$. A feasible plan would need $2 \leq x_1 + x_2 \leq 1$, which is impossible. The feasible set is empty: the problem is *infeasible*. In practice: an error in the data or in the model, or a demand that is too high.

Check yourself

Q3. In the drone workshop, the profit of a drone B becomes 80 thousand DA (and A stays at 40 thousand DA). Which case is it? Give all the optimal plans.
Q4. Classify: max $x_1 + 2x_2$ subject to $x_1 + x_2 \geq 4$, $x_1, x_2 \geq 0$.

8 Special situations

8.1 A redundant constraint

Add to the drone workshop the constraint "at most 25 drones A per week": $x_1 \leq 25$. The testing constraint already gives $3x_1 \leq 60 - 2x_2 \leq 60$, so $x_1 \leq 20$. The line $x_1 = 25$ is outside the polygon: the new constraint removes no feasible plan, and the optimum does not change. Such a constraint is *redundant*. It can be removed. On the picture, a redundant constraint has a line outside the feasible set, or touching it at only one corner.

8.2 A degenerate corner

max $2x_1 + x_2$ subject to $x_1 \leq 4$, $x_2 \leq 2$, $x_1 + x_2 \leq 6$, $x_1, x_2 \geq 0$. The feasible set is the rectangle with corners $(0, 0)$, $(4, 0)$, $(4, 2)$, $(0, 2)$. At $(4, 2)$, three constraints are binding: $x_1 = 4$, $x_2 = 2$ and $x_1 + x_2 = 6$. A corner where three or more lines meet (in the plane) is *degenerate*. The constraint $x_1 + x_2 \leq 6$ is also redundant: it only touches the rectangle at this corner. The corner method works as usual: $z = 0, 8, 10, 2$ at the four corners, so the optimum is $(4, 2)$ with $z^* = 10$. Degenerate corners will matter for the simplex method.

8.3 An equality constraint

Suppose the drone workshop must use exactly all 40 assembly hours: $x_1 + 2x_2 = 40$. The feasible plans are on this line and satisfy the other constraints. On the line, $x_1 = 40 - 2x_2$, and:

- cameras: $x_2 \leq 18$;
- testing: $3(40 - 2x_2) + 2x_2 \leq 60$, that is $120 - 4x_2 \leq 60$, so $x_2 \geq 15$.

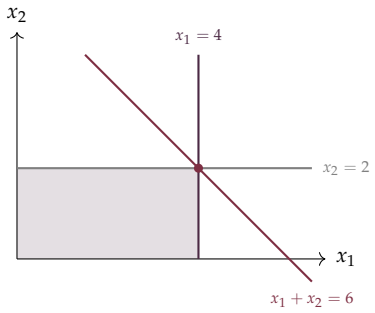


Figure 8: A degenerate corner: three lines meet at $(4, 2)$.

So the feasible set is the segment from C $= (4, 18)$ to B $= (10, 15)$. Its two ends are its corners: C gives 1240 and B gives 1300. The optimum is B. In general, an equality constraint reduces the feasible set to a line, a segment, or a single point.

9 Finding the corners without drawing

Method 2.4 — Corners by algebra

A corner is a point where two constraint lines meet *and* which satisfies all the constraints. For each pair of lines: solve the system (Method 2.1), then check all the constraints.

For the drone workshop there are 5 constraints (including $x_1 \geq 0$ and $x_2 \geq 0$), so 10 pairs:

Two lines		Point	Feasible?	z
assembly	testing	$(10, 15)$	yes	1300
assembly	cameras	$(4, 18)$	yes	1240
assembly	$x_1 = 0$	$(0, 20)$	no: 20 cameras > 18	
assembly	$x_2 = 0$	$(40, 0)$	no: testing $120 > 60$	
testing	cameras	$(8, 18)$	no: assembly $44 > 40$	
testing	$x_1 = 0$	$(0, 30)$	no: 30 cameras > 18	
testing	$x_2 = 0$	$(20, 0)$	yes	800
cameras	$x_1 = 0$	$(0, 18)$	yes	1080
cameras	$x_2 = 0$	no point (parallel lines)	—	
$x_1 = 0$	$x_2 = 0$	$(0, 0)$	yes	0

We find the five corners of the picture, and the best one is $(10, 15)$. With many variables the number of pairs (or groups) of constraints becomes enormous; this is why we will need the simplex method.

Corners and slack variables. In standard form (Lecture 1), the workshop reads $x_1 + 2x_2 + s_1 = 40$, $3x_1 + 2x_2 + s_2 = 60$, $x_2 + s_3 = 18$, with all variables ≥ 0. A constraint is binding exactly when its slack is 0. At each corner, two of the five variables are 0, one for each line through the corner:

Corner	x_1	x_2	s_1	s_2	s_3
O $(0, 0)$	0	0	40	60	18
A $(20, 0)$	20	0	20	0	18
B $(10, 15)$	10	15	0	0	3
C $(4, 18)$	4	18	0	12	0
D $(0, 18)$	0	18	4	24	0

The simplex method uses this idea: it describes a corner by the variables that are zero, and moves from a corner to a better neighbouring corner.

Walking from corner to corner. Instead of computing all the corners, we can walk along the edges (Figure 9). Start at a corner, for example O ($z = 0$). Move to a neighbouring corner (joined by an edge) with a better value: A (800), then B (1300). At B, the two neighbours A (800) and C (1240) are worse, so we stop: B is optimal. Another path, O $\to$ D (1080) $\to$ C (1240) $\to$ B, also works. This is the idea of the simplex method, studied in the next lectures.

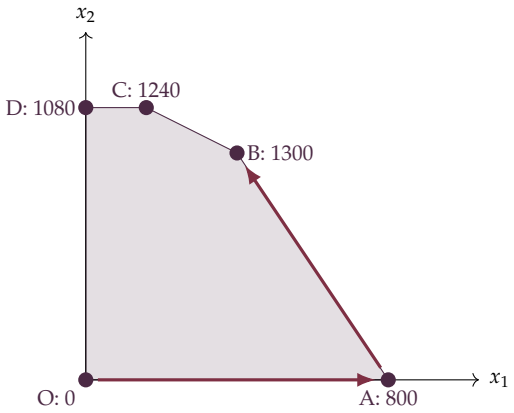


Figure 9: Walking from corner to corner: O $\to$ A $\to$ B; at B no neighbour is better.

10 What if a profit changes?

If the profit of a drone A is c_1 thousand DA (instead of 40), the values at the corners are

$$z(\mathrm{A}) = 20c_1, \quad z(\mathrm{B}) = 10c_1 + 900, \quad z(\mathrm{C}) = 4c_1 + 1080, \quad z(\mathrm{D}) = 1080, \quad z(\mathrm{O}) = 0.$$

- B is at least as good as C when $10c_1 + 900 \geq 4c_1 + 1080$, that is $6c_1 \geq 180$, so $c_1 \geq 30$.
- B is at least as good as A when $10c_1 + 900 \geq 20c_1$, that is $900 \geq 10c_1$, so $c_1 \leq 90$.
- For $c_1 \geq 30$, $z(\mathrm{B}) \geq 1200 > 1080 = z(\mathrm{D})$, and B is always better than O.

So the plan $(10, 15)$ stays optimal for every profit c_1 between 30 and 90 thousand DA. For $c_1 = 30$ or $c_1 = 90$ two corners tie (Case 2). Questions of this kind are the subject of Chapter 3 (sensitivity analysis).

11 Summary: how to recognise each case

If you see…	then
the feasible set is empty	Case 5: infeasible
feasible points $x(t)$ with $z \to \infty$ (the line can be pushed forever)	Case 4: unbounded
the last contact is one corner	Case 1: unique optimum
two neighbouring corners give the same best value	Case 2: infinitely many optima
a line outside the set, or touching it only at a corner	redundant constraint
three or more lines through one corner	degenerate corner

- Tools: draw a line with two points; choose a side with a test point; compute a corner by solving two equations (Method 2.1).
- Graphical method (Method 2.3): push the line of equal value in the direction of c (max) or $-c$ (min).

- Key fact 2.1: if an optimum exists, a corner is optimal; corner method (Method 2.2); corners by algebra (Method 2.4).
- To prove that a plan is optimal: multiply binding constraints by positive numbers and add them to get "$z \leq$ its value".

Reading. F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, Section 3.1; H. A. Taha, *Operations Research: An Introduction*, Section 2.2.

Answers to "Check yourself"

Q1. $x_1 = 0$ gives $3x_2 = 12$, $x_2 = 4$: point $(0, 4)$. $x_2 = 0$ gives $x_1 = 12$: point $(12, 0)$. At $(0, 0)$: $0 \leq 12$ is true, so $x_1 + 3x_2 \leq 12$ keeps the side of the origin.

Q2. Add the two equations: $3x_1 = 12$, so $x_1 = 4$; then $x_2 = x_1 - 2 = 2$. Check: $8 + 2 = 10$ and $4 - 2 = 2$. The point is $(4, 2)$.

Q3. $z = 40x_1 + 80x_2 = 40(x_1 + 2x_2)$: the lines of equal value are parallel to the assembly line. Corners: O: 0, A: 800, B: $400 + 1200 = 1600$, C: $160 + 1440 = 1600$, D: 1440. B and C tie: Case 2. All the points $(4 + 6t,\ 18 - 3t)$, $0 \leq t \leq 1$, are optimal, with $z^* = 1600$.

Q4. The feasible set is the region beyond the line $x_1 + x_2 = 4$; it goes to infinity. The points $(0, t)$ with $t \geq 4$ are feasible ($0 + t \geq 4$) and their value is $2t$, as large as we want. The problem is unbounded (Case 4).

Q5. On the left: $\frac{20}{3}(2x_1 + x_2) + \frac{50}{3}(x_1 + 2x_2) = \frac{40+50}{3}x_1 + \frac{20+100}{3}x_2 = 30x_1 + 40x_2$. On the right: $\frac{20}{3} \times 8 + \frac{50}{3} \times 7 = \frac{160+350}{3} = 170$. Since the multipliers are positive, every feasible plan satisfies $30x_1 + 40x_2 \geq 170$: no purchase costs less than 170 thousand DA. The plan $(3, 2)$ costs exactly 170 thousand DA, so it is optimal.